System Vision document

1. Problem
   In busy streets in downtown and Heliopolis, delivery vans, ride hailing cars, buses and cars all share the same parking slots which are limited. and the government has no data on how curb space is used
2. Vision
   For all the companies who suffer from the overcrowding and limited parking space, smart curb is a curb management platform that turns each curb into a digital, online bookable space which is time based and has enforced rules using sensors and plate recognizing cameras.
3. Traffic Planners (design zones and rules), the Head of Traffic Planning (approves rule changes), delivery companies and drivers (book slots), ride-hailing and navigation apps (consume live data), Traffic Police (need legally valid fines), shop owners (customer stops).
4. Key Capabilities
   - Zones and rules: planners draw zones on a map, set rule schedules by day and hour (rules are copyable between streets), Rule changes need the department head's approval and are published 7 days ahead, except emergencies.
   - Booking: driver can book 10, 20 or 30 minutes slots in under 30s from a phone, a booking is accepted only if the vehicle fits otherwise another zone or time is recommended.
     Companies can book recurring slots, capped at 40% of a zone in peak hours.
   - Enforcement: cameras can detect plates in every zone. Vehicles with no booking or 5+ minutes past will get fined.
     A warning first for registered drivers, then fines to the company. Private cars are fined through the traffic violations system. Drivers can dispute with photo evidence.

Data and services: a live feed for navigation and ride hailing apps (uber, didi or indrive). Ride hailing is charged per stop.